



Tokyo Electron (TSE:8035)

TP:270.58 USD (82% upside, 16.8%
IRR)

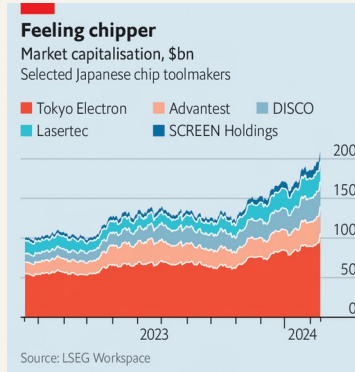
> Industry Overview

Tokyo Electron has a **7.9% market share in US** semiconductor machine manufacturing and **12% globally**. Our world is built on semiconductors — chip demand is set to rise over the coming decade. The global semiconductor manufacturing equipment market is set to grow at a **CAGR of 10.4%** from **\$91.20bn in 2023** to **\$182.30bn**.

Global market share of key competitors (the big 5)



- TEL is the 4th largest company in Japan by market cap.
- It is among the big 5 equipment manufacturers.
- The combined market cap of the 5 most valuable Japanese semiconductor manufacturers have doubled in dollar terms over the past year



Key Trends:

1. Advanced Manufacturing Technologies:

- Innovations in lithography, etching and deposition processes enabling production of smaller more powerful & energy-efficient chips.
- EUV lithography systems with higher NAs (numerical apertures) developed to allow for smaller feature sizes on chips

2. Advanced Packaging Technologies:

- System-in-Package (SiP), Fan-Out Wafer-Level Packaging (FOWLP), 3D stacking = mass production of high-performance semiconductors

3. Miniaturization & Node advancement:

- Moving toward smaller nodes (e.g. 7nm, 5nm).

➤ Industry Overview - Outperforms in every cycle

Annual Stock Returns (CAGR%)	3-year	5-year	10-year	15-year	20-year
SOX (Semi) Index	16%	26%	25%	20%	14%
Semicap Equipment	21%	31%	26%	21%	15%
Chip Design Software (EDA)	18%	31%	30%	24%	18%
SP500 Index	9%	14%	12%	12%	9%
NASDAQ Composite Index	7%	18%	16%	15%	12%

Source: Bloomberg Finance L.P. Pricing as of 10/17/2024 market close.

Suppose you are a long-term investor, **semiconductors should be a natural choice** because of:

- **Proven Outperformance:** Semiconductor, semiconductor capital equipment, and EDA stocks have demonstrated a strong multi-year track record of outperformance across 3, 5, 10, and 15-year horizons
- **Critical Role in Tech Value Chain:** Semiconductors remain integral to the technology ecosystem, driving sustained demand.
- **Structural Profitability & FCF Improvements:** Enhanced structural profitability and free cash flow generation underpin the sector's resilience.

These factors collectively position the sector for continued outperformance relative to broader market indices.

> Company Overview

Tokyo Electron (TYO: 8035) is engaged in the development, manufacturing, and sale of semiconductor production equipment and industrial electronics products for flat panel display (FPD) manufacturing equipment.

TEL is the **only company** that produces the necessary equipment for the **four key wafer front-end processes**:

Microscopic circuits are formed by repeating processes 1-4

1 Deposition:

- Metal thin films deposited on wafer surface preparatory to forming device structures/wiring



TEL offers:

Various depo systems (e.g oxide/anneal, CVD, ASFD, ALD, PVD)

2 Lithography:

- Transferring extremely fine patterns on wafer surfaces: photo-resist coating, exposure, development



TEL has:

90% share of overall coater/developer market. ~100% share for the high NA process.

3 Etching:

- Defines circuit pattern formed by lithography



2nd largest market share in dry etch. Distinctive tech in critical processes.

Q3 Capitalization

Share Price - \$147
Shares Out. - 461mn

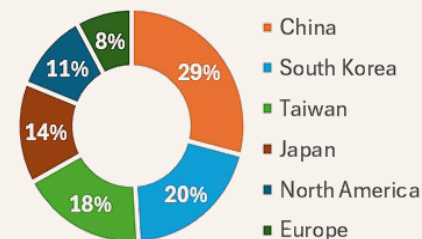
Market Cap. - \$67,731mn
(-) Cash & Equiv. - \$3,668mn
(-) Total Debt - \$0

Enterprise Value - \$64,062mn

Projected Income Statement

Year	Revenue (Y bn)	%YoY	EBIT (Y bn)	%YoY	EBIT Margin	CapEx /EBITDA	Dividend Payout
2023	2,209	10.24%	617	3.08%	27.96%	10.13%	50%
2024	1,830	-17.13%	456	-26.14%	24.93%	24.57%	50%
2025E	2,368	29.37%	666	46.04%	28.14%	23.17%	50%
2026E	2,595	9.6%	761	14.25%	29.33%	16.72%	50%
2027E	2,897	11.64%	890	16.97%	30.73%	14.5%	50%

Sales by Geography



4 Cleaning:

- Removal of foreign objects to prevent defects in process.



2nd largest market share of wafer cleaning systems in the industry.

➤ Investment Thesis

Tokyo Electron's competitive advantage cannot be eroded

Market has misunderstood company after AI exuberance; trades at **unjustified discount to peers**.

Market is discounting semiconductors but does not recognise glut is over

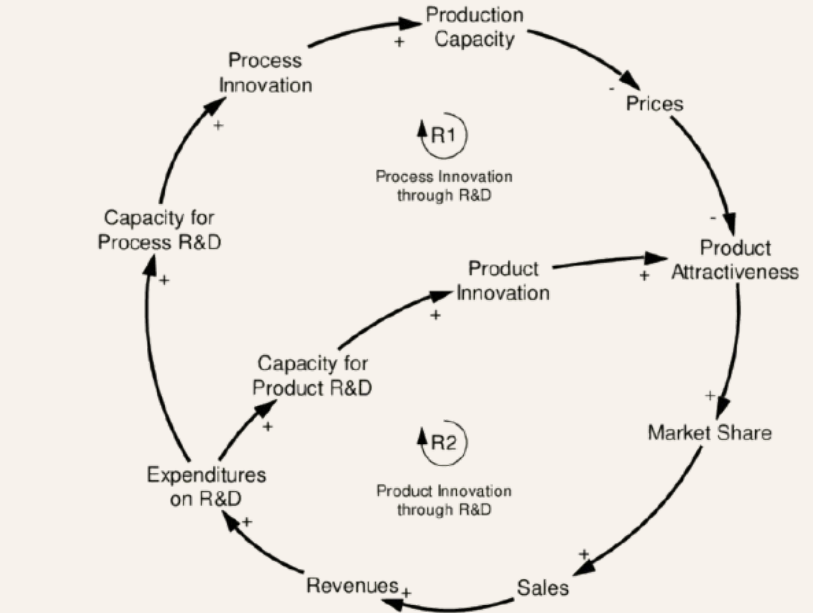
Market **pricing** semiconductor companies as though we are in a glut, yet **we are 30% into the upturn**.

Japanese strength

underestimated

Secular macro shifts, unprecedented corporate reforms and **unique environment for semiconductors**.

In the 1980s, TLE underinvested due to poor economic conditions. Korean manufacturers did the opposite and were able to penetrate the equipment market. **Management won't repeat yester-year's mistakes, R&D and Capex both set to double over previous 5 years:** Doubling R&D to \$10bn NEXT 5 years.



- ~100% employee retention accumulates knowledge.
- Sophistication of supply chain over decades

➤ 1b) Competitive advantage can not be eroded

Idiosyncratic advantages unchanged, just like its peers ASML, ATAM, the company has a durable competitive advantage and monopoly in its niche.

In fact, over time its adoption of 'Shift Left' and outsized AI exposure should cause advantages to accumulate.

Historically, traded roughly at same P/E as its peers. **Today**, it trades at a **6 turns discount** to this peer set.

Tokyo Electron is poised to have outsized benefits from AI trends, and the market over-emphasised that in the June rally.

But now it is mispriced; price correction **was too steep** after the AI rally in June.

Fiscal Year	2024A	2024E	2025E	2026E
Revenue	\$ 14,885.67	\$ 15,289.94	\$ 16,757.92	\$ 18,707.71
Revenue Growth	15.54%	26.34%	9.60%	11.64%
EBITDA	\$ 4,526.92	\$ 4,732.59	\$ 5,385.74	\$ 6,187.29
EBITDA Margin	30.41%	30.95%	32.14%	33.07%
EV/EBITDA	18.75	13.27	11.66	10.15
P/E	27.58	20.12	17.61	15.05

Sell-side agrees with our income projections but disagrees on price

Year	P/E of TOELY	Avg. P/E of Comparables	TOELY Premium
2020	21.32	23.32	-2.00
2021	29.95	28.82	1.13
2022	18.23	15.98	2.26
2023	20.42	18.64	1.77
4-yr Avg	22.48	21.69	0.79
21.11.24	21.46	28.03	-6.56

P/E premium relative to peers has narrowed but fundamentals are same

➤ 2a) The glut is over; market is mispricing

Semiconductors are a cyclical industry.

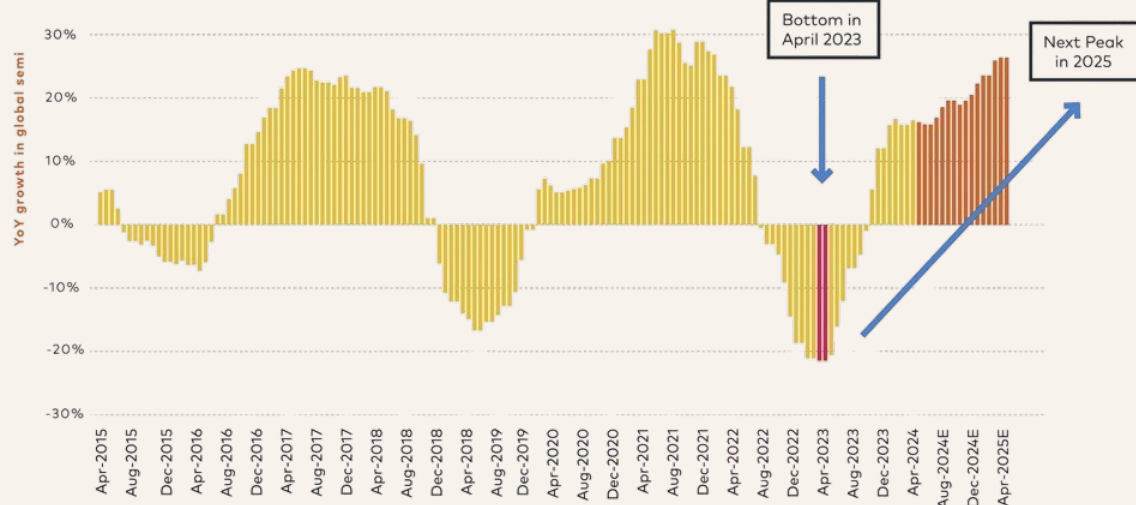
There was a **shortage period throughout 2021-22**

Through **2022-23 semiconductors have been in a glut.**

Positive growth resumes in 2024

Semicap equipment demand in turn follows semiconductor volumes at a 6-month lag

Implies **upturn in semicap equipment is now.**



Source: Jefferies Financial Group Inc. as of July, 2024 using data from World Semiconductor Trade Statistics (WSTS).

We are **30-35% into the typical 2-year upturn** phase of semis cycle

➤ 2b) The glut is over; market is mispricing

Upcycle signaled by 10 months of Y/Y growth

- Growth expected to accelerate in 2025, driven by:
 - **Edge AI adoption** (smart devices and autonomous systems)
 - **Non-AI chip restocking** after inventory depletion from prior glut.
 - **Favorable macro conditions** for global economy.

Upcycle Duration

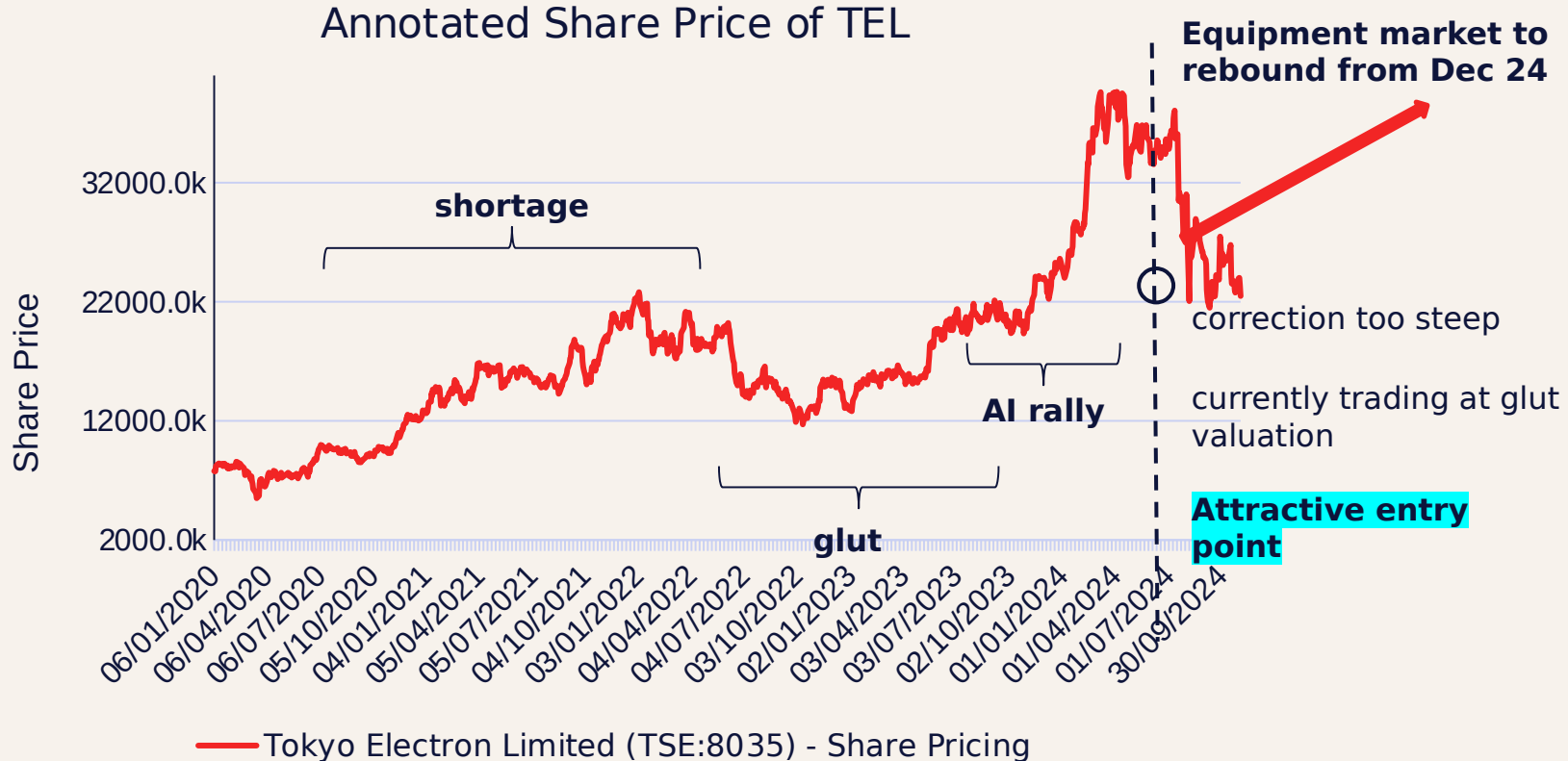
- Historical cycles last **26-30 months**
- Current upcycle is **30-35% complete**, so there is significant runway ahead

Attractive Valuation

- Sector pullback: **-18% below all-time high P/E**.
- Strong investment case in companies that sell chips for AI/datacenters
- **PEG ratio suggest undervaluation on forward earnings**



➤ 2c) The glut is over; market is mispricing



➤ 3a) Japanese strength is not being recognised

MACRO – *The economy is potentially at an inflection point*

- **Inflation = margin expansion;** it gives firms with pricing power.
- **Inflation = higher household spend;** as nominal wages can rise.
- **Inflation = lower real rates;** BOJ did not raise rates, so Yen slid.
“Land is cheap, water is cheap, people are cheap” - TSMC CEO

POLICY – *Executive support benefits the national champion*

- **Japan’s semiconductor subsidy/GDP ratio top in world** at 0.7%.
- **Corporate reforms**, notably incl. simplifying acquisition rules and NISA
- **Favorable geopolitics** – TLE exempt from U.S. sanctions, yet still benefits from its CHIPS

SECTORAL – *Japan has unique endowments*

- Japanese universities are at the **forefront of wafer fab research**.
- *E.g. OIST recently discovered a cheaper method to do **EUV lithography***
- Japanese workforce consists of the **largest pool of cheap, skilled technicians and**

YEAR	CPI
2024E	2.8%
2023	3.27%
2022	2.50%
2021	-0.23%
2020	-0.03%
2019	0.47%

➤ 3b) Japanese strength is not being recognised

— MSCI JAPAN — MSCI WORLD — MSCI USA



Japan has been **relatively undervalued on a P/E basis for over a decade.**

A decadal macro shift and/or change in Japanese saving culture would upset this norm

Note: Forward 12-month price-to-earnings ratio (P/E) of MSCI indexes are taken

Source: LSEG Eikon | Patturaja Murugaboopathy

DCF Valuation

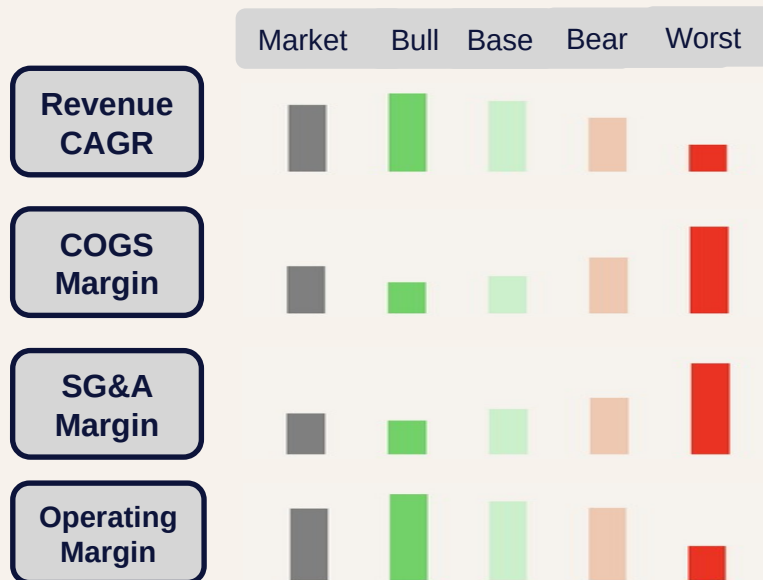
Baseline Assumptions

Summary	FY2024	FY2025	FY2026	FY2027	FY2028
Revenue					
Semiconductor Production	30.0%	28.0%	13.0%	8.0%	5.0%
Flat Panel Display Manufac	-34.8%	2.1%	8.6%	21.4%	-6.6%
Others	-53.8%	11.9%	14.0%	-3.3%	-1.7%
COGS					
SG&A	55.0%	54.0%	54.0%	54.0%	54.0%
R&D	8.5%	8.4%	8.3%	8.2%	8.2%
D&A	11.0%	11.0%	10.0%	10.0%	10.0%
Operating Margin	3.5%	3.4%	3.4%	3.3%	2.3%
Tax Rate	24.0%	23.2%	24.3%	24.5%	25.5%
	25%	25%	25%	25%	25%

Valuation Summary

 Tokyo Electron Limited	Bullish	Base	Bearish	Worst
Today's Price		148.80		
Implied Share Price	270.58	239.80	191.30	135.52
% Change	81.8%	61.2%	28.6%	-8.9%
IRR	16.8%	14.1%	9.3%	2.7%
MIRR	16.0%	13.5%	9.1%	2.9%

Assumptions Summary



Discount Rate	10%
Exit Multiple (EV/EBITDA)	15.0x
Financing Rate	10%
Reinvestment Rate	6%

Public Comparables

Public Comps	Market Data				Profitability				Financial Health			Valuation Multiples			
Company Name	Price	Enterprise Va	Market Capi	1 Yr Grow	EBITDA / NI	Margin	EBITDA / NI	Growth	D/E Rati	LT D/E R	Debt/EE	EV/Sale	EV/EBIT	EV/UFC	P/E
Unit	USD	mUSD	mUSD	%	%	%	%	%	%	%	%	x	x	x	x
Advantest Corporation (TSE:6857)	59.01	43142.8	43629.5	18.4%	28.8%	17.7%	26.5%	24.1%	19.3%	18.2%	0.5x	11.2x	38.8x	73.1x	64.0x
Lam Research Corporation (NasdaqGS:LAMR)	70.05	89048.6	90132.3	5.3%	31.6%	26.0%	-1.4%	2.1%	58.8%	52.9%	1.0x	5.7x	17.2x	21.8x	22.7x
Lasertec Corporation (TSE:6920)	117.78	10489.2	10622.7	36.7%	45.2%	29.7%	34.9%	27.9%	6.9%	-	0.1x	8.0x	17.7x	140.0x	27.3x
SCREEN Holdings Co., Ltd. (TSE:7701)	61.5	4116.4	5372.5	24.9%	22.5%	14.9%	45.9%	53.4%	1.4%	0.8%	0.0x	1.1x	5.1x	49.9x	11.2x
KLA Corporation (NasdaqGS:KLA)	614.46	84386.4	82190	19.1%	41.4%	28.9%	1.3%	-4.4%	191.8%	169.6%	1.6x	8.2x	19.6x	34.2x	28.1x
ASML Holding N.V. (ENXTAM:ASML)	669.9	262981.8	263290.1	34.7%	33.8%	26.4%	-5.7%	-9.0%	29.1%	29.1%	0.5x	9.5x	27.4x	NM	36.1x
Disco Corporation (TSE:6146)	275.03	28232.8	29804.2	21.2%	45.5%	29.1%	47.4%	40.1%	-	-	-	12.1x	26.6x	50.9x	44.1x
Applied Materials, Inc. (NasdaqGS:AMAT)	168.88	136013.4	139225.4	16.2%	30.4%	26.4%	1.1%	4.7%	32.9%	28.7%	0.8x	5.0x	16.5x	23.4x	19.6x
Rorze Corporation (TSE:6323)	12.35	2141.9	2178	34.7%	30.4%	22.8%	22.6%	35.7%	29.9%	14.3%	1.1x	2.9x	9.7x	14.1x	13.2x
Amkor Technology, Inc. (NasdaqGS:AMT)	25.19	6140.5	6212.6	8.2%	16.6%	5.7%	-8.5%	-10.0%	32.5%	25.6%	1.2x	1.0x	5.4x	19.3x	17.0x
Peer Group															
High	669.9	262981.8	263290.1	36.7%	45.5%	29.7%	47.4%	53.4%	191.8%	169.6%	1.6x	12.1x	38.8x	140.0x	64.0x
Low	12.35	2141.9	2178	5.3%	16.6%	5.7%	-8.5%	-10.0%	1.4%	0.8%	0.0x	1.0x	5.1x	14.1x	11.2x
Mean	207.42	66669.4	67265.7	21.9%	32.6%	22.8%	16.4%	16.5%	44.7%	42.4%	0.8x	6.5x	18.4x	47.4x	28.3x
Median	93.92	35687.8	36716.9	20.1%	31.0%	26.2%	12.0%	14.4%	29.9%	27.1%	0.8x	6.9x	17.5x	34.2x	25.0x
Tokyo Electron Limited (TSE:8035)	144.04	62995.5	66390.4	17.5%	31.40%	22.06%	31.62%	37.65%	10%	8.90%	0.2x	4.6x	15.0x	28.6x	22.0x

➤ Tokyo Electron - Risks and Mitigants

Delayed rebound in end-product demand

Tokyo Electron's revenue is closely tied to the cyclical nature of the semiconductor industry, which experiences boom-and-bust cycles based on global demand for chips. NB: This can be upside or downside

Tokyo Electron has focused on the development of equipment for high-growth sectors like automotive semiconductors and AI/IoT chips.

Impact



Likelihood



Geopolitical and Export Controls (* Appendix D)

US-China trade tensions and export controls on advanced semiconductor equipment can impact TEL's ability to sell products to key customers, particularly in China. NB: TLE lesser exposed to China than peers

Tokyo Electron has increased its focus on expanding its customer base in regions like Taiwan and South Korea, which house major players such as TSMC and Samsung.

Impact



Likelihood



Rapid Technological Advancements and Competition

The semiconductor equipment industry is highly competitive, with companies like ASML, Applied Materials, and Lam Research innovating rapidly. Falling behind in technology can erode TEL's market position.

Aggressive R&D; Tokyo Electron has made very significant investments in extreme ultraviolet (EUV) lithography-compatible processes.

Impact



Likelihood



> Catalyst Path



Near-Term Catalysts

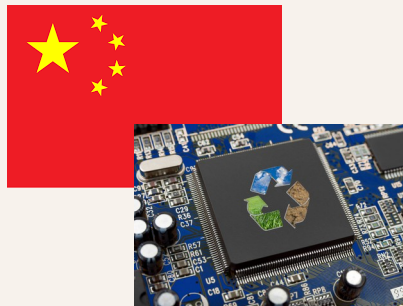
Record CapEx Spending; TSMC announced \$32B for 2024

US CHIPS Act; \$52B in US subsidies

EUV Momentum; ASML projects EUV shipments to be up by 30% in 2024

Japan's Domestic Investment; Rapidus (Japan's national semiconductor venture) begins building a \$37 B fab

FY 2025



FY 2026

China's Mature Node Expansion; China's semiconductor fab spending expected to exceed \$17B

Sustainability Push; TEL commits to delivering 10% more energy-efficient equipment to meet the sustainability requirements of TSMC and Samsung fabs

Market Share Expansion; TEL targets 25% global market share in wafer cleaning tools, a \$10 billion market

Global Semiconductor Revenue Growth; Semiconductor revenue projected to surpass \$1 trillion by 2030, doubling TEL's addressable market

Long-Term Catalysts



➤ Appendix A. Supplementary Valuation Materials

Period		Actual FY2019	Actual FY2020	Actual FY2021	Actual FY2022	Actual FY2023	Projected FY2024	Projected FY2025	Projected FY2026	Projected FY2027	Projected FY2028	Net Interest I % of Revenue	mUSD %	10.38 0.097%	7.03 0.054%	12.97 0.077%	18.32 0.108%	16.97 0.137%	16.68 0.123%	15.85 0.108%	17.49 0.115%	17.51 0.112%	18.26 0.114%	
Revenue												Income/(Los	mUSD -	10.04 22.33	14.17 22.40	22.33 (27.40)	22.40 (35.10)	20.78 (12.70)	21.30 (3.10)	25.40 (2.30)	27.10 (1.30)	15.90 1.30		
Semiconduc	mUSD	9,860.38	11,891.61	16,007.40	16,212.60	11,302.26	12,997.60	14,037.40	14,458.53	14,892.28	15,339.0	Currency Ex	mUSD	23.60 37.44	(28.50) 24.20	(20.20) 13.21	(10.50) 26.02	(27.40) 33.80	(35.10) 30.10	(12.70) 24.34	3.10 27.22	(2.30) 25.78	1.30 26.50	
% Growth	%		20.6%	34.6%	1.3%	-30.3%	15.0%	8.0%	3.0%	3.0%	3.0	Other Non-C	mUSD	37.44 (4.20)	24.20 (12.00)	13.21 0.58	26.02 0.82	33.80 66.33	30.10 76.87	24.34 38.84	27.22 54.20	25.78 5.60	26.50 7.30	
Flat Panel Di	mUSD	614.23	757.44	492.70	565.86	570.64	372.13	379.87	412.50	500.97	467.9	Asset Writed	mUSD -	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	- (3.30)	
% Growth	%		23.3%	-35.0%	14.9%	0.8%	-34.8%	2.1%	8.6%	21.4%	-6.6	Other Unusu	mUSD	(0.40) (33.80)	(33.80) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	(42.00) (42.00)	
Others	mUSD	424.35	471.96	539.46	419.48	476.40	219.98	246.23	280.76	271.61	267.1	Tax	mUSD	552.21 25.0%	669.95 23.1%	1,314.48 26.6%	1,152.99 24.8%	723.78 24.8%	822.43 25.0%	748.69 25.0%	808.06 25.0%	849.82 25.0%	872.02 25.0%	
% Growth	%		11.2%	14.3%	-22.2%	13.6%	-53.8%	11.9%	14.0%	-3.3%	-1.7	Tax Rate	%	25.0%	23.1%	26.6%	24.8%	24.8%	25.0%	25.0%	25.0%	25.0%	25.0%	
Unallocated	mUSD	(179.30)	(198.50)	(230.60)	(256.30)	-	-	-	-	-	-	Net Income	mUSD	1,721.21 16.1%	2,196.59 17.0%	3,599.29 21.4%	3,547.50 20.9%	2,406.29 19.5%	3,305.50 24.3%	3,002.00 20.5%	3,270.23 21.6%	3,386.38 21.6%	3,467.81 21.6%	
% Growth	%		10.7%	16.2%	11.1%	-100.0%	0.0%	0.0%	0.0%	0.0%	0.0	NI Margin	%	16.1%	17.0%	21.4%	20.9%	19.5%	24.3%	20.5%	21.6%	21.6%	21.6%	
Total Revenue	mUSD	10,719.66	12,922.51	16,808.96	16,941.64	12,349.30	13,589.71	14,663.50	15,151.79	15,664.86	16,074.0	DCF	mUSD	2,205.27 552.21	2,899.52 669.95	4,934.93 1,314.48	4,646.83 1,152.99	2,917.56 723.78	3,289.74 822.43	2,994.76 748.69	3,232.24 808.06	3,399.28 849.82	3,488.08 872.02	
% Growth	%		20.5%	30.1%	0.8%	-27.1%	10.0%	7.9%	3.3%	3.4%	2.6	Tax	mUSD	552.21 243.23	669.95 272.30	1,314.48 307.80	1,152.99 324.20	723.78 346.00	822.43 475.64	748.69 659.86	808.06 681.83	849.82 626.59	872.02 642.96	
COGS	mUSD	6,276.31	7,542.17	8,992.40	9,212.22	6,613.05	7,474.34	8,064.93	8,333.48	8,615.67	8,840.7	Change in N'	mUSD	(407.80) 458.8	1,220.47 486.5	1,584.14 538.68	519.93 572.88	(145.20) 773.5	(386.14) 977.20	312.31 439.91	348.98 340.31	275.60 419.95	340.10 321.48	
% of Revenue	%	58.5%	58.4%	53.5%	54.4%	53.6%	55.0%	55.0%	55.0%	55.0%	55.0	CapEx	mUSD	458.8 458.8	486.5 486.5	538.68 538.68	572.88 572.88	773.5 773.5	977.20 977.20	439.91 439.91	340.31 340.31	419.95 419.95	321.48 321.48	
Gross Profit	mUSD	4,443.35	5,380.33	7,816.56	7,729.42	5,736.25	6,115.37	6,598.58	6,818.30	7,049.19	7,233.3	Unlevered F	mUSD	1,845.29 61,965.62	794.90 61,965.62	1,805.44 61,965.62	2,725.23 61,965.62	1,911.48 61,965.62	2,351.88 61,965.62	2,453.71 61,965.62	2,666.72 61,965.62	2,480.51 61,965.62	3,097.44 61,965.62	
Gross Margi	%	41.5%	41.6%	46.5%	45.6%	46.4%	45.0%	45.0%	45.0%	45.0%	45.0	PV of Exit Ve	mUSD	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88	61,965.62 75,015.88
SG&A	mUSD	877.13	972.98	1,270.61	1,320.11	1,131.42	1,155.13	1,330.98	1,389.05	1,456.83	1,494.8	Implied EV	mUSD	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9	75,015.88 12552.9
% of Revenue	%	8.2%	7.5%	7.6%	7.8%	9.2%	8.5%	9.1%	9.2%	9.3%	9.3	Total Debt	mUSD	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98	12552.9 62,462.98
R&D	mUSD	1,117.71	1,235.53	1,303.23	1,438.28	1,341.26	1,494.87	1,612.99	1,515.18	1,566.49	1,607.41	Equity Value	mUSD	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9	62,462.98 460.9
% of Revenue	%	10.4%	9.6%	7.8%	8.5%	10.9%	11.0%	11.0%	10.0%	10.0%	10.0	Number of S	m	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	460.9 135.52	
EBITDA	mUSD	2,448.50	3,171.82	5,242.73	4,971.03	3,263.56	3,465.38	3,654.61	3,914.08	4,025.87	4,131.04	Implied Shai	USD	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	135.52 135.52	
EBITDA Marg	%	22.8%	24.5%	31.2%	29.3%	26.4%	25.5%	24.9%	25.8%	25.7%	25.7%													
D&A	mUSD	243.23	272.30	307.80	324.20	346.00	475.64	659.86	681.83	626.59	642.96													
% of Revenue	%	2.3%	2.1%	1.8%	1.9%	2.8%	3.5%	4.5%	4.5%	4.0%	4.0%													
Operating In	mUSD	2,205.27	2,899.52	4,934.93	4,646.83	2,917.56	3,289.74	2,994.76	3,232.24	3,399.28	3,488.08													
Operating M	%	20.6%	22.4%	29.4%	27.4%	23.6%	24.2%	20.4%	21.3%	21.7%	21.7%													

➤ Appendix B. Supplementary Valuation Materials

1 Bullish	FY2024	FY2025	FY2026	FY2027	FY2028		
Revenue						CAGR	18.6%
Semiconductor Production	35.0%	30.0%	13.0%	10.0%	5.0%		53.4%
COGS	55.0%	53.0%	53.0%	53.0%	53.0%		8.2%
SG&A	8.5%	8.4%	8.2%	8.1%	8.0%		10.4%
R&D	11.0%	11.0%	10.0%	10.0%	10.0%		3.5%
D&A	3.5%	3.5%	3.5%	3.5%	3.5%		
Operating Margin							

2 Base	FY2024	FY2025	FY2026	FY2027	FY2028		
Revenue						CAGR	16.8%
Semiconductor Production	30.0%	28.0%	13.0%	8.0%	5.0%		54.2%
COGS	55.0%	54.0%	54.0%	54.0%	54.0%		8.3%
SG&A	8.5%	8.4%	8.3%	8.2%	8.2%		10.4%
R&D	11.0%	11.0%	10.0%	10.0%	10.0%		3.2%
D&A	3.5%	3.4%	3.4%	3.3%	2.3%		
Operating Margin							

3 Bearish	FY2024	FY2025	FY2026	FY2027	FY2028		
Revenue						CAGR	12.8%
Semiconductor Production	25.0%	20.0%	8.0%	6.0%	5.0%		55.0%
COGS	55.0%	55.0%	55.0%	55.0%	55.0%		8.4%
SG&A	8.5%	8.5%	8.4%	8.4%	8.3%		10.4%
R&D	11.0%	11.0%	10.0%	10.0%	10.0%		
D&A	3.0%	2.8%	2.7%	2.5%	2.3%		
Operating Margin							

4 Worst	FY2024	FY2025	FY2026	FY2027	FY2028		
Revenue						CAGR	6.4%
Semiconductor Production	15.0%	8.0%	3.0%	3.0%	3.0%		55.0%
COGS	55.0%	55.0%	55.0%	55.0%	55.0%		9.1%
SG&A	8.5%	9.1%	9.2%	9.3%	9.3%		10.4%
R&D	11.0%	11.0%	10.0%	10.0%	10.0%		
D&A	3.5%	4.5%	4.5%	4.0%	4.0%		
Operating Margin							

Public Comps	Market Data				Profitability				Financial Health				Valuation Multiples				
Company Name	Price	Enterprise Va	Market Capi	1 Yr Grow	EBITDA % NI	Margin	EBITDA % NI	Growth	D/E	Rati LT	D/E	Rz	Debt/EE	EV/Sale	EV/EBIT	EV/UFC	P/E
Unit	USD	mUSD	mUSD	%	%	%	%	%	%	%	%	%	x	x	x	x	x
Advantest Corporation (TSE:6857)	59.01	43142.8	43629.5	18.4%	28.8%	17.7%	26.5%	24.1%	19.3%	18.2%	0.5x	11.2x	38.8x	73.1x	64.0x		
Lam Research Corporation (Nasdaq: LSCX)	70.05	89048.6	90132.3	5.3%	31.6%	26.0%	-1.4%	2.1%	58.8%	52.9%	1.0x	5.7x	17.2x	21.8x	22.7x		
Lasertec Corporation (TSE:6920)	117.78	10489.2	10622.7	36.7%	45.2%	29.7%	34.9%	27.9%	6.9%	-	0.1x	8.0x	17.7x	140.0x	27.3x		
SCREEN Holdings Co., Ltd. (TSE:7733)	61.5	4116.4	5372.5	24.9%	22.5%	14.9%	45.9%	53.4%	1.4%	0.8%	0.0x	1.1x	5.1x	49.9x	11.2x		
KLA Corporation (NasdaqGS:KLA)	614.46	84386.4	82190	19.1%	41.4%	28.9%	1.3%	-4.4%	191.8%	169.6%	1.6x	8.2x	19.6x	34.2x	28.1x		
ASML Holding N.V. (ENXTAM:ASX)	669.9	262981.8	263290.1	34.7%	33.8%	26.4%	-5.7%	-9.0%	29.1%	29.1%	0.5x	9.5x	27.4x	NM	36.1x		
Disco Corporation (TSE:6146)	275.03	28232.8	29804.2	21.2%	45.5%	29.1%	47.4%	40.1%	-	-	-	12.1x	26.6x	50.9x	44.1x		
Applied Materials, Inc. (NasdaqGS:A	168.88	136013.4	139225.4	16.2%	30.4%	26.4%	1.1%	4.7%	32.9%	28.7%	0.8x	5.0x	16.5x	23.4x	19.6x		
Rorze Corporation (TSE:6323)	12.35	2141.9	2178	34.7%	30.4%	22.8%	22.6%	35.7%	29.9%	14.3%	1.1x	2.9x	9.7x	14.1x	13.2x		
Amkor Technology, Inc. (NasdaqGS:AMK)	25.19	6140.5	6212.6	8.2%	16.6%	5.7%	-8.5%	-10.0%	32.5%	25.6%	1.2x	1.0x	5.4x	19.3x	17.0x		
Peer Group																	
High	669.9	262981.8	263290.1	36.7%	45.5%	29.7%	47.4%	53.4%	191.8%	169.6%	1.6x	12.1x	38.8x	140.0x	64.0x		
Low	12.35	2141.9	2178	5.3%	16.6%	5.7%	-8.5%	-10.0%	1.4%	0.8%	0.0x	1.0x	5.1x	14.1x	11.2x		
Mean	207.42	66669.4	67265.7	21.9%	32.6%	22.8%	16.4%	16.5%	44.7%	42.4%	0.8x	6.5x	18.4x	47.4x	28.3x		
Median	93.92	35687.8	36716.9	20.1%	31.0%	26.2%	12.0%	14.4%	29.9%	27.1%	0.8x	6.9x	17.5x	34.2x	25.0x		
Tokyo Electron Limited (TSE:803)	144.04	62995.5	66390.4	17.5%	31.40%	22.06%	31.62%	37.65%	10%	8.90%	0.2x	4.6x	15.0x	28.6x	22.0x		

Comments

Margins were average level compared with peers

Share price and valuation multiples fall on the lower end of the spectrum

This implies TEL is undervalued compared with its peer group and offers a large upside potential if margins improve as projected

#1
#2
#3

➤ Appendix C. Peer Set Selection

COMPARABLE COMPANY ANALYSIS TOKYO ELECTRON (TSE: 80035)

[Applied Materials, Inc. \(NasdaqGS: AMAT\)](#) *

[Lam Research Corporation \(NasdaqGS: LRCX\)](#) *

[Advantest Corporation \(TSE: 6857\)](#)

[ASML Holding N.V. \(ENXTAM: ASML\)](#) *

[Amkor Technology, Inc. \(NasdaqGS: AMKR\)](#)

[Rorze Corporation \(TSE: 6323\)](#)

[KLA Corporation \(NasdaqGS: KLAC\)](#) *

[Disco Corporation \(TSE: 6146\)](#)

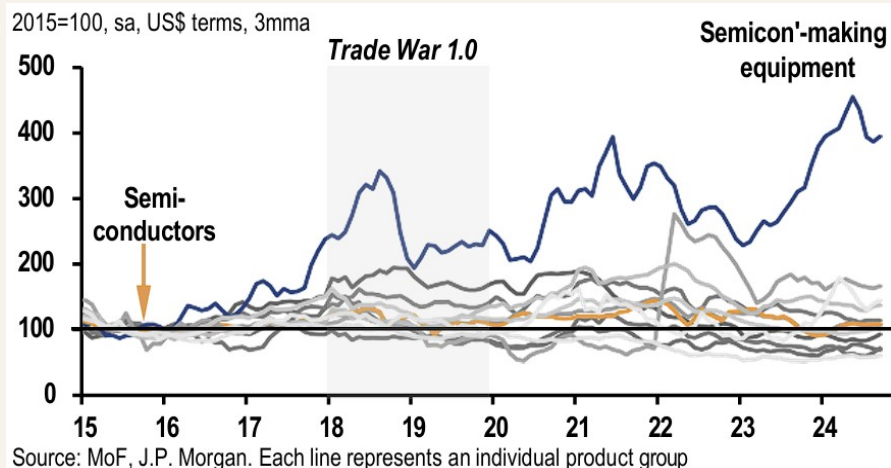
[Lasertec Corporation \(TSE: 6920\)](#)

[SCREEN Holdings Co., Ltd. \(TSE: 7735\)](#)

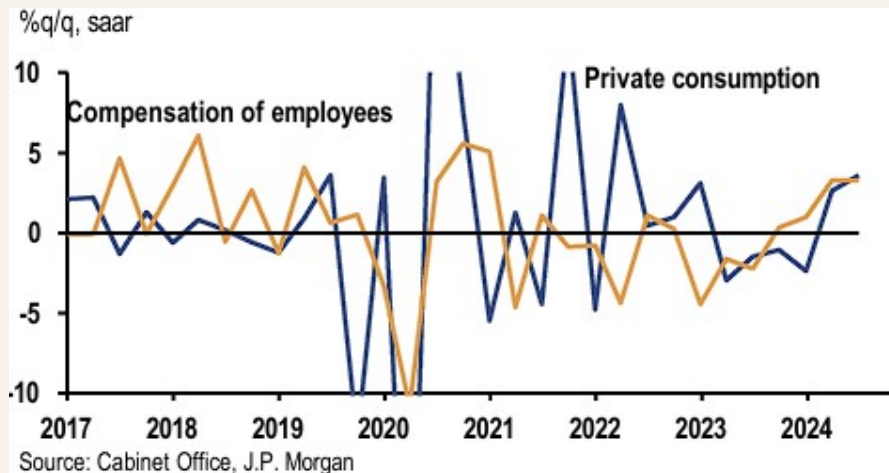
[Tokyo Electron Limited \(TSE: 8035\)](#) *

* = Big 5 SME

➤ Appendix D. Geopolitical Restrictions and Consumption



China appears already to be front-loading imports of Japanese equipment in advance of potential restrictions. (See the differential between the blue and black line).



Last quarter's 3.6%q/q saar rise in private consumption marks one of the strongest quarters for real household spending over the past decade (excl. pandemic swings)

Disclaimer

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